

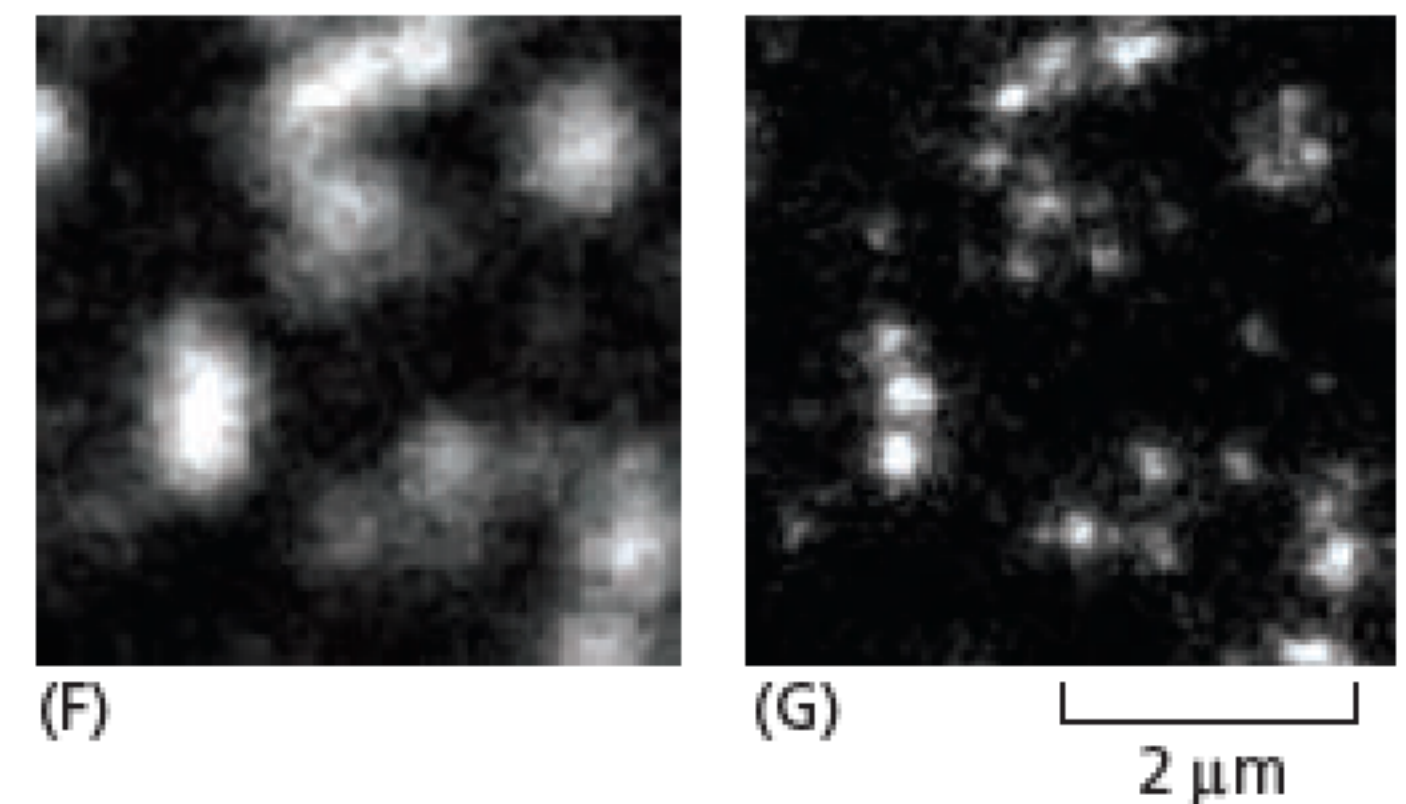
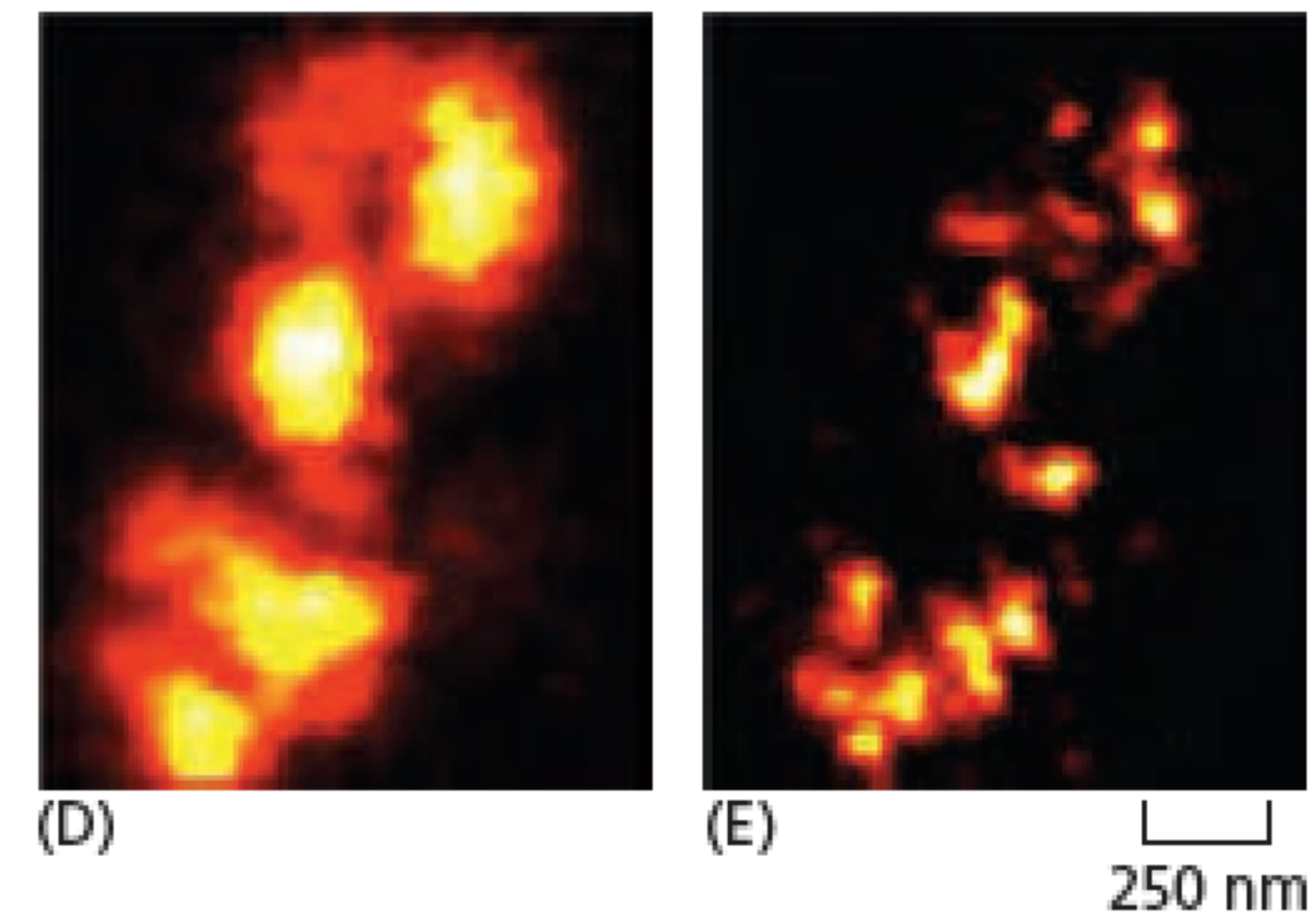
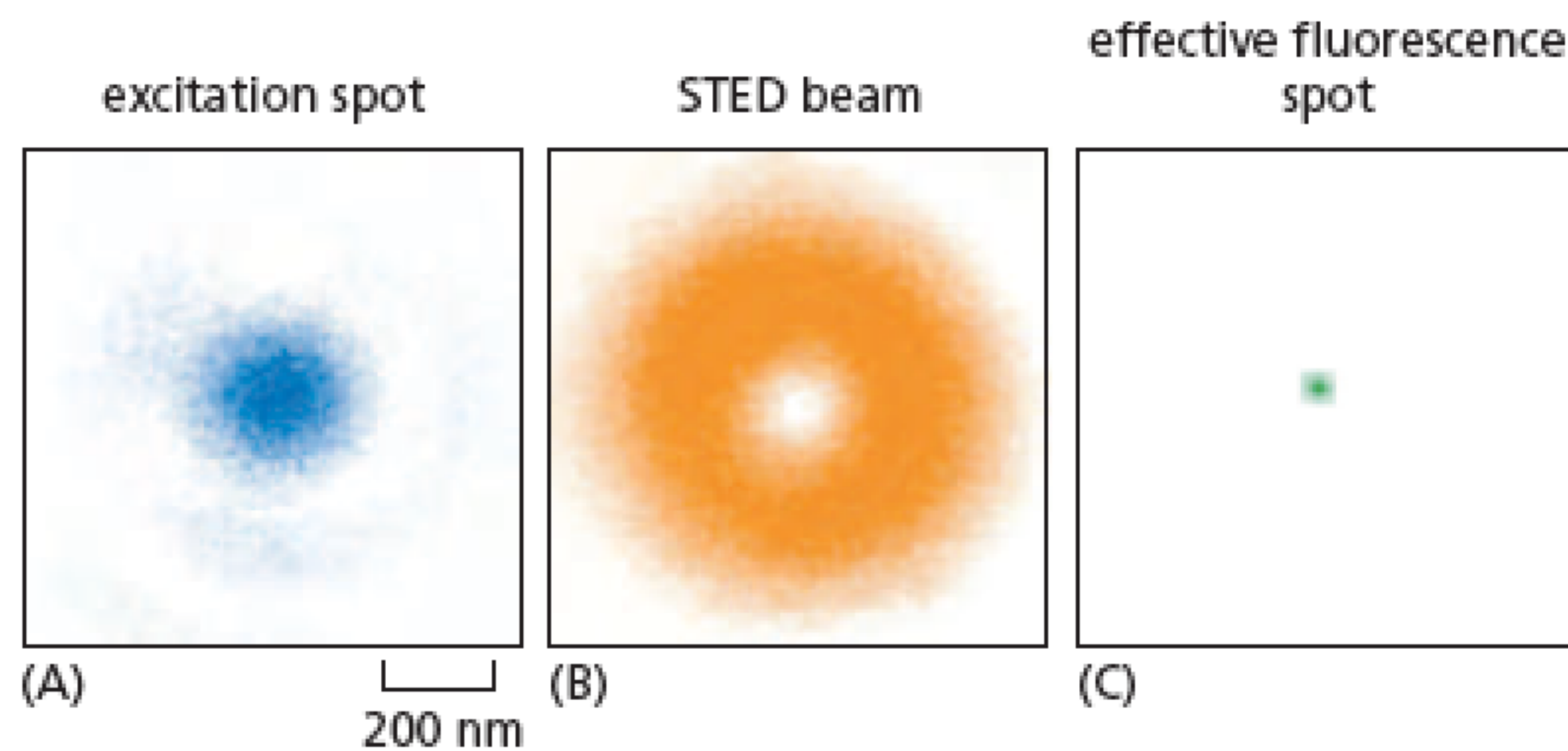


Figure 9–37 Superresolution microscopy can be achieved by reducing the size of the point spread function. (A) The size of a normal focused beam of excitatory light. (B) An extremely strong superimposed laser beam, at a different wavelength and in the shape of a torus, depletes emitted fluorescence everywhere in the specimen except right in the center of the beam, reducing the effective width of the point spread function (C). As the specimen is scanned, this small point spread function can then build up a crisp image in a process called STED (stimulated emission depletion microscopy). (D) **Synaptic vesicles in live cultured neurons**, fluorescently labeled and imaged by ordinary confocal microscopy, with a resolution of 260 nm. (E) The same vesicles imaged by STED, with a resolution of 60 nm, which allows single vesicles to be resolved. (F) **Fluorescently labeled replication factories in the nucleus** of a cultured cell, imaged by ordinary confocal microscopy. (G) The same replication factories imaged by STED. Single, discrete replication sites can be resolved by STED that cannot be seen in the confocal image. (A, B, and C, from G. Donnert et al., *Proc. Natl Acad. Sci. USA* 103:11440–11445, 2006. With permission from National Academy of Sciences; D and E, from V. Westphal et al., *Science* 320:246–249, 2008. With permission from AAAS; F and G, from Z. Cseresnyes, U. Schwarz and C.M. Green, *BMC Cell Biol.* 10:88, 2009.)